Aggregation as a Routing Cofunctor

Extracted from a machine-checked development

Abstract

We construct a routing policy from the concrete eleven-facility Sawasdee Cellars wine network to a two-node aggregate network (Australia to Thailand) and verify that it is a morphism of directed containers. The shape map sends each facility to its geographic region; the position map lifts each aggregate route to a canonical concrete route out of the given facility. Shape compatibility, root preservation, and shift compatibility are established by finite case analysis. The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

Aggregation in logistics replaces a fine-grained network of facilities by a coarse network of regions, while retaining enough route structure that policies expressed at the regional level can be interpreted concretely. In the language of directed containers, such a policy is a morphism from the concrete network to the aggregate network: a map on shapes (facilities to regions) together with a *lift* of aggregate positions to concrete positions, compatible with roots, sub-shapes, and shifts.

The present note isolates one fully explicit instance. Starting from the Sawasdee Cellars wine network $\mathcal{N}_{\text{wine}}$ (eleven facilities, ten shipping legs), we form a two-node aggregate digraph G_{agg} with regions {australia, thailand} and a single aggregate leg. A routing policy

$$\varphi \;:\; \mathcal{N}_{\text{wine}} \;\Longrightarrow\; \text{network}(G_{\text{agg}})$$

is defined by sending each facility to its region and lifting the aggregate stay-put and Australia-to-Thailand routes to canonical concrete routes. The three morphism laws are checked by exhaustive case analysis on the finite set of facilities.

The development works throughout in a single universe of types. Proofs are indicated by the phrase "by lean," meaning they have been checked by a proof assistant rather than written out in prose.

What this note claims. In ordinary mathematical language: coarsening the wine network to two geographic regions extends to a directed-container morphism. Regional route data lift canonically to facility-level routes, and the lift preserves the algebraic structure of roots and path concatenation.

2 Directed-container morphisms

We recall the data of a morphism of directed containers in the form used below. Let $\mathcal{C}$ and $\mathcal{D}$ be directed containers, with shape types $S_{\mathcal{C}}$ and $S_{\mathcal{D}}$, position families $P_{\mathcal{C}}$ and $P_{\mathcal{D}}$, and structure maps root, sub, shift.

Definition 2.1 (Directed-container morphism)**.** A **morphism of directed containers** $\varphi : \mathcal{C} \Rightarrow \mathcal{D}$ consists of

1. a map on shapes $u : S_{\mathcal{C}} \to S_{\mathcal{D}}$, and
2. for each shape $s \in S_{\mathcal{C}}$, a map on positions

$$\ell_s \;:\; P_{\mathcal{D}}(u(s)) \;\longrightarrow\; P_{\mathcal{C}}(s)$$

(a *lift* of $\mathcal{D}$-positions to $\mathcal{C}$-positions),

subject to three laws:

(M0) *Shape compatibility.* For every s and every position $\alpha \in P_{\mathcal{D}}\big(u(s)\big)$,

$$u\Big(\mathrm{sub}_{\mathcal{C}}\big(s,\, \ell_s(\alpha)\big)\Big) \;=\; \mathrm{sub}_{\mathcal{D}}\big(u(s),\, \alpha\big).$$

(M1) *Root preservation.* For every s,

$$\ell_s\big(\mathrm{root}_{\mathcal{D}}(u(s))\big) \;=\; \mathrm{root}_{\mathcal{C}}(s).$$

(M2) *Shift compatibility.* For every s, every $\alpha \in P_{\mathcal{D}}\big(u(s)\big)$, and every $\kappa \in P_{\mathcal{D}}\big(\mathrm{sub}_{\mathcal{D}}(u(s), \alpha)\big)$,

$$\ell_s\big(\mathrm{shift}_{\mathcal{D}}\big(u(s),\, \alpha,\, \kappa\big)\big) \;=\; \mathrm{shift}_{\mathcal{C}}\Big(s,\; \ell_s(\alpha),\; \ell_{\mathrm{sub}_{\mathcal{C}}(s,\ell_s(\alpha))}(\kappa')\Big),$$

where κ' is κ transported along the equality (M0).

Remark 2.2. The variance on positions is covariant for directed-container morphisms in this formulation (lifts of target positions to source positions), matching the cofunctorial reading of aggregation: coarse routes are refined to fine routes.

3 The aggregate network

Definition 3.1 (Aggregate digraph)**.** Let G_{agg} be the directed graph with

1. nodes {australia, thailand}, and
2. a single edge au2th : australia $\to$ thailand.

Write $\mathcal{N}_{\mathrm{agg}} := \mathrm{network}(G_{\mathrm{agg}})$ for the associated directed container.

Remark 3.2 (Aggregate positions)**.** The positions of $\mathcal{N}_{\text{agg}}$ are elementary:

- at australia: the empty route $\langle$australia, nil$\rangle$ and the single-leg route $\langle$thailand, cons(au2th, nil)$\rangle$;
- at thailand: only the empty route $\langle$thailand, nil$\rangle$.

4 The concrete wine network and the region map

Recall the Sawasdee Cellars network $\mathcal{N}_{\text{wine}} = \text{network}(G_{\text{wine}})$: eleven facilities and ten shipping legs from vineyards in South Australia through an export agency and sea freight to Thai distributors and retail shops.

Definition 4.1 (Region assignment)**.** Define a map region : Facility $\to$ {australia, thailand} by

$$\begin{aligned}
\text{region}(\text{vineyard}_i) &= \text{australia} && (i = 1, 2, 3),\\
\text{region}(\text{sawasdee}) &= \text{australia}, &&\\
\text{region}(\text{melbournePort}) &= \text{australia}, &&\\
\text{region}(\text{thaiPort}) &= \text{thailand}, &&\\
\text{region}(\text{distributor}_j) &= \text{thailand} && (j = 1, 2),\\
\text{region}(\text{retail}_k) &= \text{thailand} && (k = 1, 2, 3).
\end{aligned}$$

Thus the five Australian-side facilities map to australia and the six Thai-side facilities map to thailand. The sea leg of the concrete network is the unique edge that crosses the regional boundary.

5 Lifting aggregate routes

Write $\text{AgPos}(r)$ for $\text{Pos}(\mathcal{N}_{\text{agg}})(r)$ and $\text{Route}(f)$ for $\text{Pos}(\mathcal{N}_{\text{wine}})(f)$.

Definition 5.1 (Position lift)**.** Define, for each facility f, a map

$$\ell_f \;:\; \text{AgPos}\big(\text{region}(f)\big) \;\longrightarrow\; \text{Route}(f) \tag{1}$$

by the following canonical choices.

1. **Australian facilities.** If f is Australian and the aggregate route is the empty route at australia, lift to the empty concrete route at f. If the aggregate route ends in thailand, lift to the unique canonical concrete route from f to thaiPort along the spine

   $$f \;\to\; \cdots \;\to\; \text{melbournePort} \xrightarrow{\text{sea}} \text{thaiPort}$$

   (prefixing the legs $v_i s$, *smp*, and/or sea as required by the facility). Concretely:

   - from vineyard_i: legs $v_i s$, *smp*, sea;
   - from sawasdee: legs *smp*, sea;
   - from melbournePort: leg sea.

2. **Thai facilities.** If f is Thai, the only aggregate position is the empty route at thailand; lift it to the empty concrete route at f.

Remark 5.2. Because G_{wine} is a polytree along the Australia-to-Thailand spine used here, the "canonical" concrete route from each Australian facility to Thai Port is unambiguous. The lift does not attempt to choose among downstream distributor or retail alternatives; those live entirely within the Thai region and are invisible to the two-node aggregate.

6 The morphism laws

Proposition 6.1 (Shape compatibility, M0)**.** *For every facility f and every aggregate position $\alpha \in \mathrm{AgPos}(\mathrm{region}(f))$,*

$$\mathrm{region}\Big(\mathrm{sub}_{\mathcal{N}_{\mathrm{wine}}}(f,\, \ell_f(\alpha))\Big) \;=\; \mathrm{sub}_{\mathcal{N}_{\mathrm{agg}}}(\mathrm{region}(f),\, \alpha). \tag{2}$$

Proof. By lean. □

Proposition 6.2 (Root preservation, M1)**.** *For every facility f,*

$$\ell_f(\mathrm{root}_{\mathcal{N}_{\mathrm{agg}}}(\mathrm{region}(f))) \;=\; \mathrm{root}_{\mathcal{N}_{\mathrm{wine}}}(f). \tag{3}$$

Proof. By lean. □

Proposition 6.3 (Shift compatibility, M2)**.** *For every facility f, every aggregate position $\alpha \in \mathrm{AgPos}(\mathrm{region}(f))$, and every aggregate continuation $\kappa \in \mathrm{AgPos}(\mathrm{sub}_{\mathcal{N}_{\mathrm{agg}}}(\mathrm{region}(f), \alpha))$, the lift of the aggregate shift equals the concrete shift of the lifted positions (with the continuation transported along Proposition 6.1).*

Proof. By lean. □

Remark 6.4 (Idea of the case analysis)**.** All three laws are proved by matching on the finite type of facilities and on the (at most two) aggregate positions at the image region. Representative situations:

- Empty aggregate route at an Australian facility: sub-shape stays Australian; lifting the root is the concrete root; shifting by a further Australia-to-Thailand leg reproduces the canonical concrete crossing.
- Australia-to-Thailand aggregate route: sub-shape is Thailand; the concrete lift ends at Thai Port; further aggregate continuations are necessarily the Thai empty route, and concatenating the empty suffix leaves the concrete route unchanged.
- Thai facilities: only empty routes appear, and all structure maps are trivial.

Theorem 6.5 (Routing policy)**.** *The pair* (region, ℓ) *is a morphism of directed containers*

$$\varphi \;:\; \mathcal{N}_{\mathrm{wine}} \;\Longrightarrow\; \mathcal{N}_{\mathrm{agg}}. \tag{4}$$

Proof. By lean. □

7 Discussion

What has been established. Geographic aggregation of the Sawasdee Cellars network to two regions extends to a directed-container morphism. The construction supplies a fully checked instance of aggregation-as-cofunctor: coarse route algebra is interpreted by lifting to fine routes along a fixed regional policy.

On canonicity. The position lift chooses, for each Australian facility, a single spine route to Thai Port. Alternative concrete realisations (e.g. different downstream branching inside Thailand) are invisible at the two-node aggregate level; a finer aggregate would require a richer lift.

Scope. The development treats one concrete pair of networks and does not develop a general theory of aggregation morphisms for arbitrary partitions of nodes. It also does not address the interaction of routing policies with compositional trackings (e.g. whether monoid-valued data descends along φ).

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content follows the aggregation construction in the supply-chain development that motivated this note. The presentation is a prose rendering of a formal development intended for human review of the theorem statements.

References

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